DRIVEPULSE CAR RENTALS

Production System Architecture & Operational Workflow Specification

1. SYSTEM OVERVIEW & BUSINESS ARCHITECTURE

DrivePulse is a full-stack, enterprise-grade Car Rental Management web application designed specifically for administrative, management, and operational staff. It completely digitalizes fleet dispatch, customer verification, pricing negotiation, digital vehicle inspections, payments, statutory Indian GST compliance, and executive analytics.

Key Operational Tenets

- **Staff-Only Access:** Strict Role-Based Access Control (Super Admin, Operations Manager, and Counter Staff). No public client booking interface.
- **Dynamic People-Based Pricing:** Rigid daily rates are removed. Staff manually enter negotiated rental tariffs based on passenger count and route requirements.
- **Frictionless Customer Walk-ins:** Counter staff can type customer name and phone on the fly without forced pre-registration.
- **Odometer Clutter Elimination:** Odometer metrics are captured strictly during physical vehicle handover and return inspections, keeping general fleet overviews clean.
- **Road Legality Safety Gates:** Automated document compliance engine prevents vehicle assignment if Insurance, Fitness, or PUC certificates have expired, requiring documented Manager override.
- **Statutory GST Compliance:** 18% GST (9% CGST + 9% SGST), HSN/SAC code 996601, and official downloadable PDF invoices and receipts.

2. TECHNOLOGY STACK

- **Frontend Client:** React 18, Vite, TypeScript, Tailwind CSS, Lucide Icons, React Hook Form, and Zod validation running on port 3000.
- **Backend REST API:** Node.js, Express, TypeScript, Zod schema validation, Helmet security headers, bcrypt password hashing, and rate limiting running on port 5000.
- **Database Layer:** Prisma ORM with SQLite for instant local development and 1-click migration path to PostgreSQL for enterprise production.
- **Authentication:** JWT access and refresh tokens stored securely in httpOnly cookies with automatic token renewal and session timeout.
- **Document Generation:** PDFKit vector graphics engine generating signed rental agreements, GST tax invoices, and official payment receipts.

3. ROLE-BASED ACCESS CONTROL (RBAC) MATRIX

- **Super Admin:** Full unrestricted authority over all operations, users, fleet, system settings, and security audit logs.
- **Operations Manager:** Full fleet, booking, and inspection authority; can authorize compliance overrides and blacklist customers. Read-only access to system settings.
- **Counter Staff:** Operational dispatch authority: create bookings, record inspections, collect payments, and enter walk-in customers. Cannot override compliance or modify fleet.

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End-to-End Operational Workflows & Compliance Gates

4. OPERATIONAL WORKFLOW LIFECYCLE

Step 1: Walk-In / Phone Customer Booking Dispatch

When a customer contacts the rental business, counter staff navigate to /bookings/new. Step 1 offers a one-click toggle between "Enter Manually" and "Select Existing". Staff enter Full Name, Phone, and optional DL/City. In Step 3, fleet vehicles display passenger capacity (e.g. 7 Seater) with "Custom Tariff". In Step 4, staff enter the Agreed Rental Amount (¹) negotiated based on passenger count. Live quotation calculates 18% GST, Advance Paid, and Balance Due in real time.

Step 2: Fleet Road Legality & Compliance Gates

Every candidate vehicle is checked against mandatory legal certificates: Comprehensive Insurance, Commercial Fitness, and PUC. If any certificate is expired, the vehicle is flagged as blocked. Counter staff cannot dispatch a blocked vehicle. A Manager can review the block and authorize a bypass only with a documented reason, which is immutably logged into the audit trail.

Step 3: Vehicle Dispatch & Pickup Digital Inspection

At handover, yard staff perform a digital condition check. Staff record physical odometer, fuel level (0/4 to 4/4), exterior scratches/dents, photos, and handed-over accessories (Fastag, spare tyre, toolkit). The hirer signs directly on a digital touchscreen canvas. Upon submission, vehicle status transitions to RENTED, booking transitions to PICKED_UP, and an official signed PDF Rental Agreement is generated.

Step 4: Vehicle Return, Damage Settlement & Deposit Refund

Upon vehicle return, staff record return odometer and fuel level. The system prevents return odometer from being lower than pickup. Staff assess late fees, fuel deficiency charges, or new damage costs if applicable. The customer signs the return checklist. The system automatically reconciles the refundable security deposit against damages and updates vehicle status to AVAILABLE.

Step 5: Billing, Invoicing & Financial Settlement

Every transaction generates an official receipt (REC-YYYY-XXXX). Staff can generate statutory GST Tax Invoices (INV-YYYY-XXXX) under SAC 996601 with CGST and SGST splits. Official PDF invoices and receipts are downloadable at any time.

5. RELATIONAL DATA MODEL (18 PRISMA ENTITIES)

- **Core Domain:** User, Vehicle, VehicleDocument, VehicleDocumentHistory, MaintenanceRecord, Expense
- **Customer Domain:** Customer, CustomerDocument, CustomerBlacklistHistory
- **Booking & Operations:** Booking, BookingExtra, InspectionChecklist, Payment, Invoice
- **Security & Governance:** AuditLog, SystemSetting, RateCard, RateCardSeason